Zoo501 final solved paper

Define vestibular apparatus? 2marks

Vertebrate Organ of Equilibrium

• Called as vestibular apparatus, located in the inner ear. It has:

1. • Saccule
2. • Utricle
3. • Semicircular canals

2. Abnormalities due to GH?5

Disturbances in the secretion of GH lead to abnormal growth and development in humans:

Gigantism

• Hypersecretion of GH before puberty causes excessive size and stature.

Developmental Abnormalities Due to GH

Acromegaly

• Hyper secretion of GH after maturity causes enlargement of bones of the head and limbs.

Dwarfism

• Insufficient secretion of GH during childhood and adolescence causes underdevelopment of The body (short stature).

3. Messines corpuscales and paccin corpuscales difference?3

4. Oxytocin? 3

Is a type of harmone . oxytocin are released by axon terminals in the posterior pituitary.
Stimulates milk ejection, uterine contractions during child birth.
Neurohypophyseal hormones are synthesized and packaged in the cell bodies of two groups of neurosecretory cells in the anterior portion of the hypothalamus:

1. Supraoptic nuclei
2. Paraventricular nuclei

5. Five types of hormone kuch as type ka q th?

Based on their structure and pathway for synthesis, hormones are divided into four groups:

- Peptide and protein hormones
- Amine hormones
- Steroid hormones
- Prostaglandins

6. Gastrointestinal track? 5

7. Prostaglandins sa related q th? 10 mrks
baki kuch yad ni.

Prostaglandins are about 16 hormone-like local regulators substances .

- They are derived from cyclic, long-chain, unsaturated, hydroxyl fatty acids.

Prostaglandins are produced by all or nearly all tissues.

- They are synthesized in membranes from arachidonic acid, which is produced by cleavage of membrane phospholipids by phospholipases.

In some cases, they act locally as paracrine agents.

- In other cases they act on distant target tissues in endocrine fashion.
- They have a rapid, shortlasting effect, similar to that of lipid-insoluble hormones.
- They bind to cell-surface receptors linked to the cAMP pathway.

Functions

- Prostaglandins have diverse actions on a variety of tissues, particularly involving smooth muscles.

Prostaglandins Role in Fertilization •

Prostaglandins present in semen aid in fertilization by reacting with the female cervical mucus to make it more receptive to sperm movement.

- They also stimulate the smooth muscles of the uterine wall to contract, helping sperms reach an egg.

Prostaglandins Role in Immune Response

- Damaged tissues produce prostaglandins during immune response.
- Some prostaglandins act as local regulators of inflammation.
- They promote fever and inflammation and also intensify the sensation of pain.

They worsen pain by increasing nociceptor sensitivity.

- Prostaglandin E2, formed from interleukin-1, produced in immuneresponse, causes fever by acting on the hypothalamus to elicit fever reaction.

Prostaglandins Role in

Blood Clot Formation

• Prostaglandins help platelets to aggregate and form blood clots.
• This is the basis of use of aspirin by patients at risk for a heart attack due to formation of clots.

Maintenance of Stomach Lining

• Prostaglandins help to maintain a protectivelining in the stomach.
• As aspirin interferes with prostaglandin synthesis, long-term aspirin therapy can result in damage to this protective lining, causing stomach irritation and ulcer.

Production of Erythropoietin

• Several prostaglandins stimulate the production of erythropoietin by kidneys.
• Erythropoietin is a hormone that stimulates the production of erythrocytes.

Blood Pressure Regulation

• Some prostaglandins act on the smooth muscles of blood vessels.
• They regulate blood pressure by vasodilation and vasoconstriction.

2 examples of -ve feedback mech.

Example 1;

Breakdown of sugar in the cells generates ATP.
• Excess ATP "feeds back" and inhibits an enzyme near the beginning of pathway.
• ATP production stops.

Negative Feedback Example-2

• Control of blood sugar (glucose) by insulin.
• Receptors sense a rise in blood sugar.
• Control center (pancreas) secretes insulin.
• Insulin lowers blood sugar levels.
• As blood sugar level reaches homeostasis, pancreas stops releasing insulin.

neuromodulation?

A physiological process by which neurotransmitters released by a small group of neurons diffuse through large areas of the CNS and interact and modulate the effect of many neurons simultaneously.
• This is in contrast to the normal synaptic transmission.
Neurotransmitters involved in altering the cellular properties and efficacy of synaptic transmission of multiple postsynaptic neurons.
• They act through Gprotein linked receptors that use voltage-gated ion channels for ion movements.
• Generation of response is slow, but lasts longer.

role of corpus leutum in pregnancy

If the egg is fertilized and implants, active corpus luteum is maintained.
• So progesterone and estrogen secretion continues until placenta fully develops to take over the production of these hormones.
• At this stage, corpus luteum degenerates. In many mammals, (e.g. rat) corpus luteum is stimulated by prolactin and continues to grow and secrete estrogen and Progesterone throughout the gestation period.

Types of stimuli

Two types:
• External stimuli Stimuli of odor, touch, light, sound and gravitation
• Internal stimuli Stimuli of pain, homeostatic imbalances and blood pressure

taste buds and taste receptors

The gustatory organs ofvertebrates are called taste buds.
A taste bud is composed of about 50 types of modified epithelial cells, including:
• Supporting cells (sustentacular cells)
• Basal cells
• Taste receptor cells
• basal cells are progenitor cells that give rise to new
• They regularly generate new sensory taste receptor cells which have an active life of only 10 days.

Feedback Mechanism (2)

Mechanism of self-regulation of Biological processes. Output or product itself regulates the process Maintenance of homeostasis

Acclimization with example (2)

Adrenal cortex hormone (3)

Parts of eye take part in focusing (3)

The parts of the eye involved in focusing and image formation are:
• Cornea, a biconvex lens, pupil and retina.

Exocrine hormones with example (5)

Exocrine Secretions

• Refer to a substance thatis released via a duct through the gland to the external or internal epithelial surfaces.
• e.g. digestive secretions, milk, tears, perspiration, fluid containing sperms.

Epinephrine

Stimulate fight-or-flight response; increase blood glucose levels; increase metabolic activities

sensory adaptation example

Decrease in perceived intensity, when the intensity of the stimulus has not itself changed.
• It is due to reduction in the frequency of sensory response during a sustained stimulus.

Examples

• Stimulation of touch receptors due to cloths.
• Receptors adapt to touch input from garments.
• So they can easily detect any new touch stimulus even at locations covered by clothing.

More Examples

• Stimuli of smell
• Hot or cold water
• Dim or bright light

hormone involves in water r ion regulation

These hormones include:

• Antidiuretic hormone (ADH)
• Aldosterone
• Atrial natriuretic Hormone
• Calcitonin
• Parathormone

Role of T3 and T4.10 marks
Five hormones related to digestive system5 marks
Acclimatization2 marks

An adaptive change within an individual animal.
- Happens due to chronic exposure to new, naturally occurring environmental conditions.

Rhodopsin 2 marks

The visual pigment in the rods is rhodopsin. Rhodopsins absorb photons of light energy and produce a generator potential.
Opsins are coupled to light-absorbing photopigment Retinal.
- This form a functional pigment molecule rhodopsin.

2.define line coding

Each receptor subtype for taste sensations is connected to a particular set of axons in the nerve.
- In such an arrangement, information about one taste e.g. "sweetness" would be carried by some specific subset of axons.
- Such a pattern is called labeled line coding.

3. hormones of ovary
4.name mechanism of homeostasis and example of homeostasis

To keep internal fluctuations in a
narrow range.
- The ability to protect internal environment from the harms of fluctuations in external environment.
- Homeostasis does not mean to keep a fixed internal environment.

Mechanism of Homeostasis
- Variables
- Set point
- Living control systems VS physical control systems
- Three components: Receptors, Control centre, effectors

Homeostasis—Examples
- Homeostatic regulation of water
- Homeostatic regulation of temperature
- Regulation of pH
- Regulation of glucose concentration
- Regulation of osmotic pressure
- Regulation of oxygen level
- Regulation of ion conc.

5. Explain the types of hair cells

Hair cells are ciliary cells.
• Found in several sensory organs of vertebrates.
• They are highly sensitive mechanoreceptors.
• Responsible for transducing mechanical stimuli into electrical signals.

Systems Based on Hair Cells: Examples

• Lateral-line system of fishes and amphibians, involved in detection of motion in the surrounding water.
• This system is based on hair cells.

Hair Cell Cilia

• Many cilia project from the apical end of each cell (reason of naming).
• Cilia are of two types:
• Kinocilium
• Stereocilia
• A hair cell has a single kinocilium.
• Kinocilium has a "9 + 2" arrangement of internal microtubules (similar to other motile cilia).
• Each hair cell has 20-300nonmotile stereocilia.
• Stereocilia are structurally and developmentally distinct from the kinocilium.
• They are not formed of microtubules.
• They are formed of actin filaments.
• The stereocilia are arranged in order of increasing length from one side of the cell to the other.

6. Explain photoreceptor

Photoreceptors possess light-sensitive carotenoid pigments retinal and 3- dehydroretinal.
• Carotenoids are associated with opsin proteins to form rhodopsins.
• Rhodopsins absorb photons of light energy and produce a generator potential.

Photoreceptive Structures

• Complexity and arrangement of photoreceptors vary within animal kingdom.

Eyespot or Stigma

Eyecups or Ocelli

Eyes

Pigments in Visual Transduction

Opsins

Rhodopsin

Photopigments

7.axon

Long and thick process with constant diameter.
• Arises from cell body. Lengths vary from few mm to more than a meter.

Axon—Functions

• Specialized to conduct signals away from the cell body.
• Carry information for long distances with high fidelity and without loss.

Axon—Structural Components

• Axons contain a jelly-like semi-fluid substance Axoplasm.
• A plasma membrane called as Axolemma.
• Have mitochondria and ER but lack Nissl's granules.

Axon Terminals

• Branched terminal ends of axons.
• At each terminal are small extensions telodendria with enlarged ends terminal knobs.
• Terminal knobs produce neurotransmitters.
• Axon terminals allow transmission of signals to other neurons, glands or muscle fibers.

Axon Hillock

• At the junction of cell body and axon is a coneshaped region called axon hillock.
• Here signals are generated that travel down the axon.

Myelin Sheath

• A fatty myelin sheath cover the axons.
• It is secreted by Schwann cells.
• Myelin sheath is not continuous.
• There are nonmyelinated points called Nodes of Ranvier.

Saltatory Impulse

• Impulses jump over node to node.
• Such impulses are known as saltatory impulses.
• This pattern in insulated axons increases the speed of nerve impulse.

diseases by hypothyroidism

Characterized by low production of throxine.
• It results from the lack of dietary iodine.
• It is represented by two types of diseases:
• Cretinism
• Goiter
• Cretinism results from iodine deficiency during early stages of development. In cretinism somatic, neural and sexual development is severely retarded, metabolic rate and resistance to infection is also reduced.
• Inadequate production of thyroxine in adults leads to goiter. In goiter, TSH is produced excessively that causes overstimulation of thyroid gland resulting in its enlargement (hypertrophy).

Active potential phase

role of Fibre Zonula and ciliary muscles (3)

Fibers of zonula held the lens in place.

• They change the shape of lens by exerting an outward tension on perimeter.
• Attached with fibers of zonula are the ciliary muscles
• They contract to pull fibers of zonula, which flatten the lens. This focuses distant objects on retina.
• When they relax, lens becomes rounded and near objects are focused.

Name glandular cells of thyriod and name their hormones too. (3mqx)

Glandular Cells and Hormones of Adenohypophysis

1. Somatotropes—produce growth hormone (GH)
2. Corticotropes—produce adrenocorticotropic hormone (ACTH)
3. Thyrotropes—produce thyroid-stimulating hormone (TSH)
4. Gonadotropes—produce gonadotropic hormones (luteinizing hormone, LH and follicle stimulating hormone, FSH)
5. Lactotropes—produce prolactin (PRL)

Range fractionation

In a sense organ, sensory receptors are arranged in an order of increasing sensitivity to different range of intensities of the stimulus.

• This hierarchical arrangement of receptors is known as range fractionation. Each individual receptor, in a sense organ, covers only a fraction of the total dynamic range of the stimulus intensities.
• Receptors work together in a hierarchical way to provide discrimination of stimulus intensities.

Placental hormone

In pregnancy, placenta secretes large quantities of hormones, essential for maintenance of pregnancy.
• These hormones include:
• Human chorionic gonadotropin
• Estrogens
• Progesterone
• Human chorionic somatomammotropin

Progesteron

Progesterone is secreted in tremendous quantities by placenta.

• This secretion is about 10-fold more than normal levels. Causes development of decidual cells in endometrium. These cells play an important role in the nutrition of the early embryo.
• It decreases contractility of the uterus to prevent spontaneous abortion.

5) 2 examples of interceptive receptors 3 marks

8) 5 hormones released by gastrointestinal track with functions 5 marks
9) functions of T3 and T4 and also hyperthyroidism 10 marks

Excessive secretion of thyroid hormones as in Graves' disease.
• It is an autoimmune disorder in which antibodies that mimic TSH bind to the receptors for TSH and cause sustained thyroxine production.
It leads to:
• high body temp.
• profuse sweating
• weight loss
• Irritability
• high blood pressure
• protruding eyes (exophthalmia) Characterized by low production of throxine.
• It results from the lack of dietary iodine.
• It is represented by two types of diseases:
• Cretinism
• Goiter
• Cretinism results from iodine deficiency during early stages of development. In cretinism somatic, neural and sexual development is severely retarded, metabolic rate and resistance to infection is also reduced.
• Inadequate production of thyroxine in adults leads to goiter. In goiter, TSH is produced excessively that causes overstimulation of thyroid gland resulting in its enlargement (hypertrophy).

Stigma and ocelli 3

Eyespot or Stigma
• Simplest photoreceptive structure.
• Found in some protozoa e.g. euglena.
• It is a bright red colored organelle and has carotenoid pigments.
• It gives a sense of light and dark.
• Helps in phototaxis.

Ocelli
• Multicellular photoreceptive structures that consist of a cuplike depression containing photoreceptor cells.
• Found in cnidarians and flatworms e.g. Planaria.
• Cannot form image.
• Provide the animal a sense of direction only

Colorblindness 3

Color blindness is caused due to a mutation in one of the cone opsin genes that results in the absence of one type of pigmented cones.

- A person missing a single type of color receptive cones is unable to distinguish some colors.

Donnan equilibrium

“If diffusible solutes are separated by a membrane that is freely permeable to water and electrolytes but totally impermeable to one species of ion, the diffusible solutes become unequally distributed between the two compartments”.

Donnan equilibrium is characterized by a reciprocal distribution of the anions and cations.

- This happens because the gradients are not only chemical but are also electrical.
- There must be electro-neutrality within both compartments. i.e. positive charges must equal negative charges.